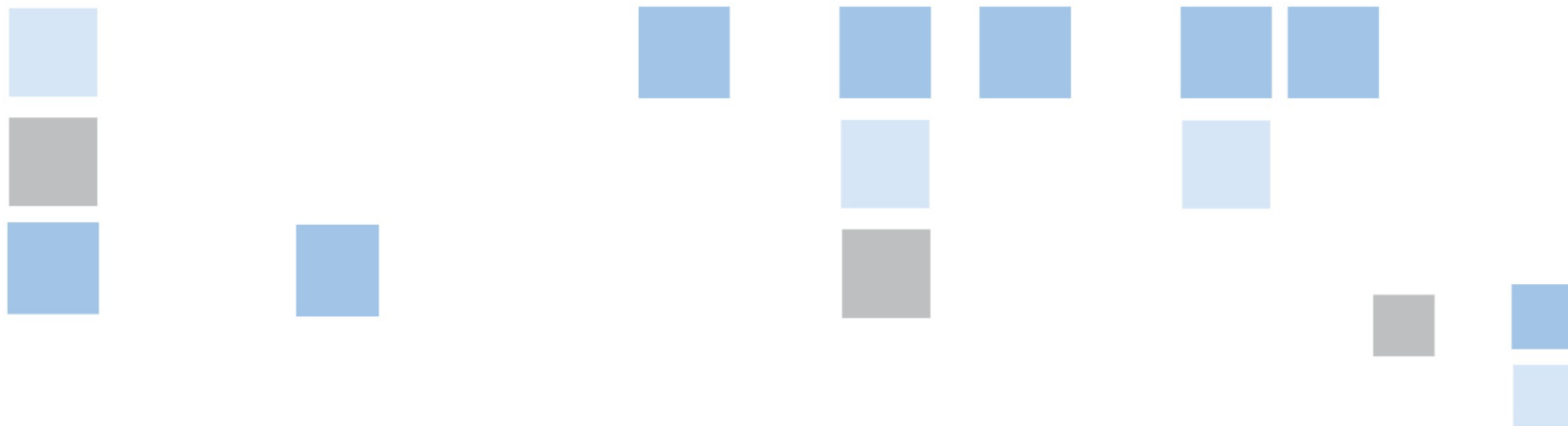
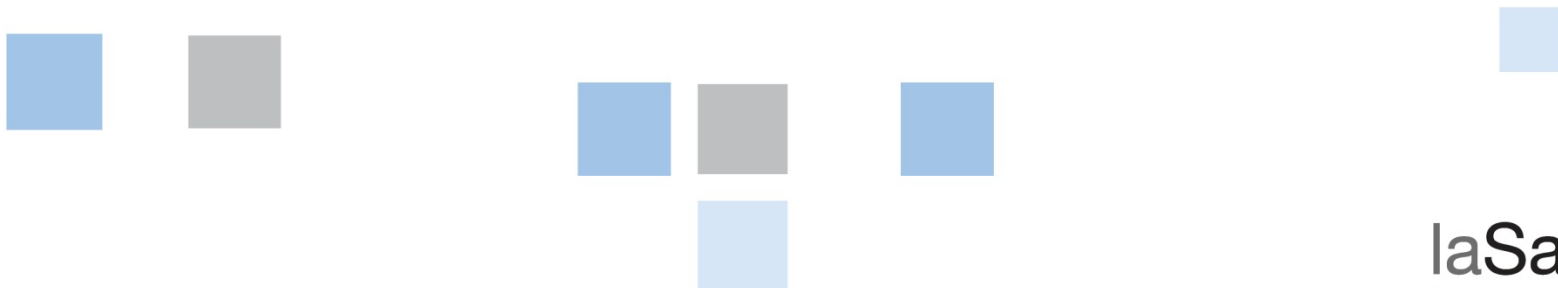




LCD

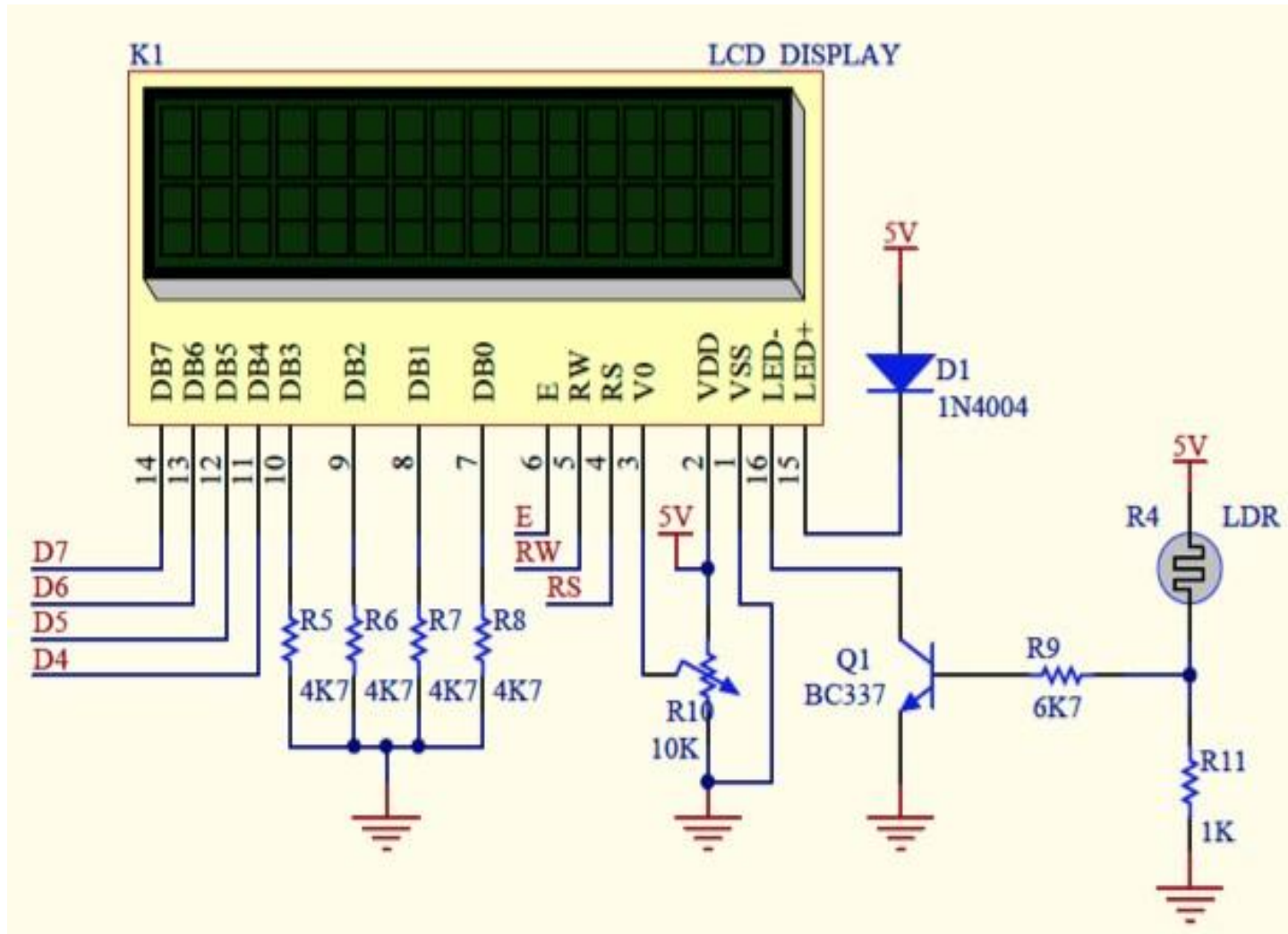


How does the LCD work?

1. Configuration and initialization stage.
2. Write the data to be displayed.



Pinout



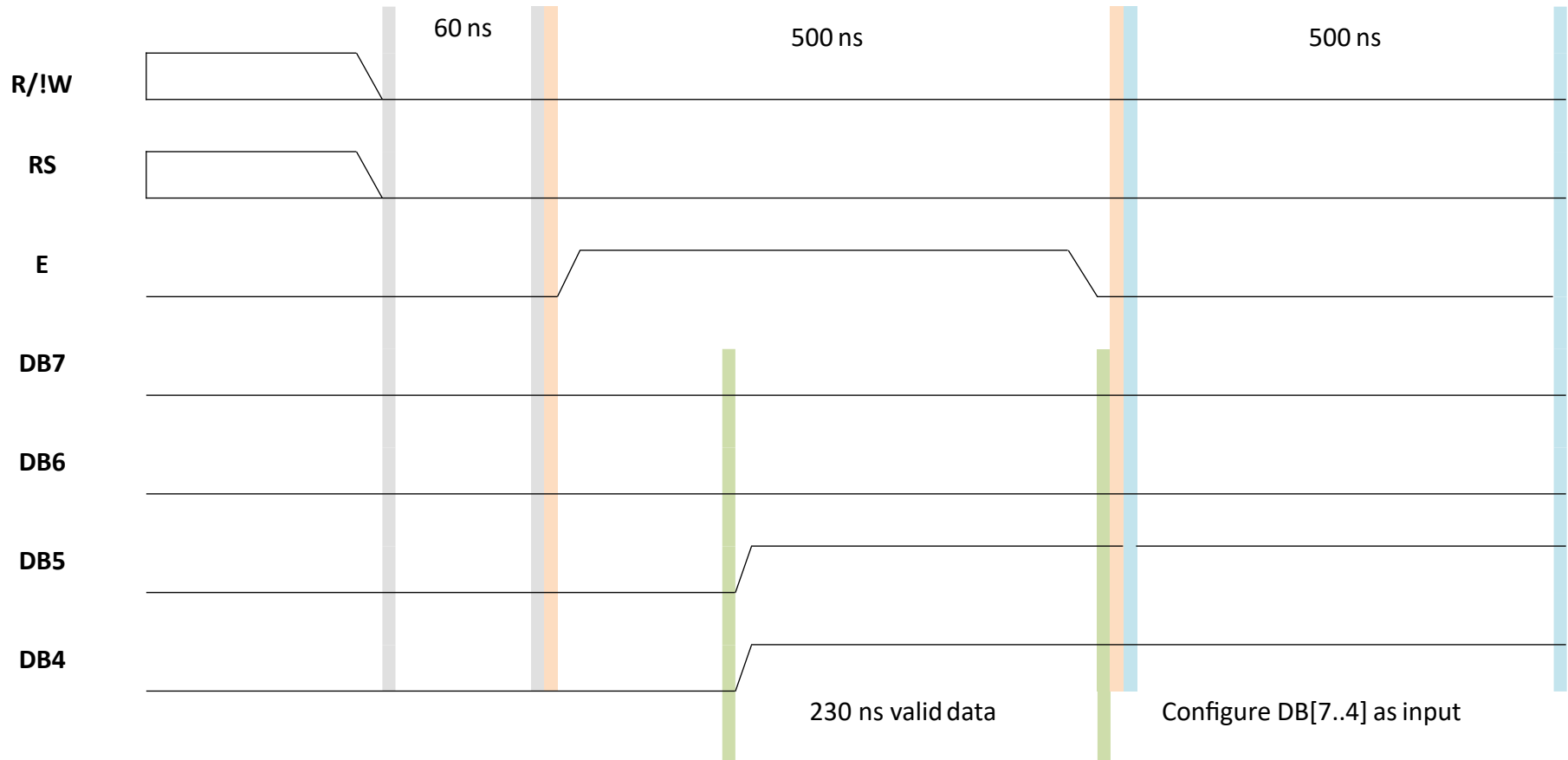
Steps

For each data or instruction:

1. Set R/!W and RS with the desired value
2. Wait > 60 ns.
3. Set Enable to 1.
4. Wait > 230 ns.
5. Place the bits DB7..DB4.
6. Wait > 230 ns.
7. Clear the Enable.

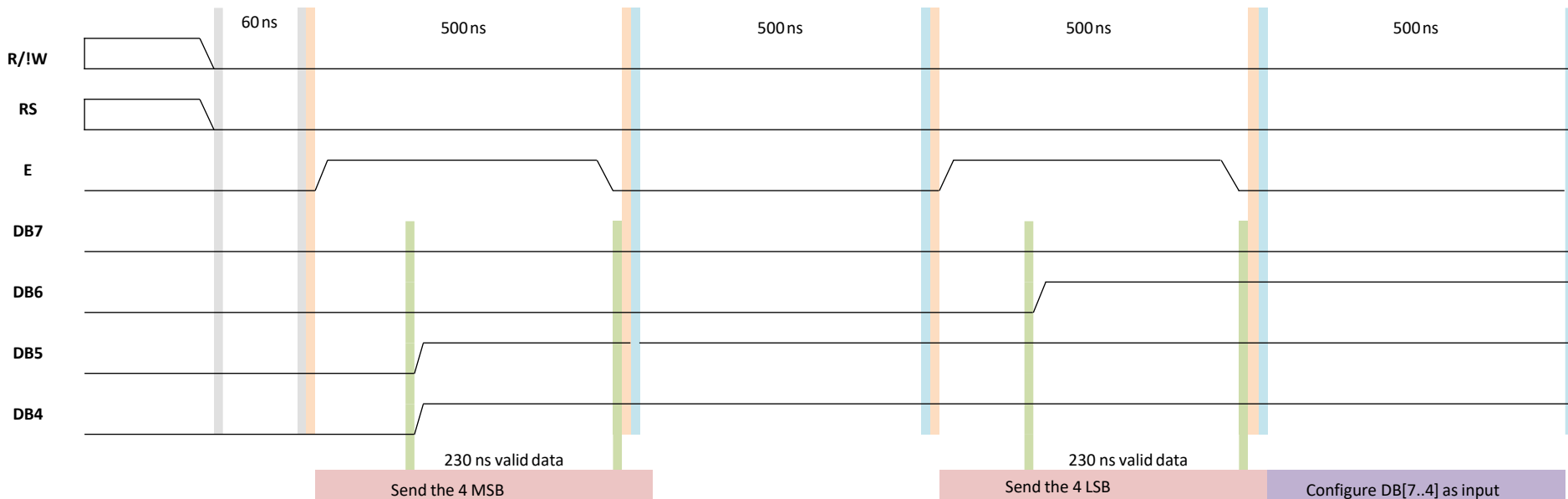
RS	R/!W	Operation
0	0	To write instructions to the LCD
0	1	To read the Busy flag (DB7)
1	0	To write our data to the LCD

Writing a 4-bits instruction:



RS	R/!W	DB7	DB6	DB5	DB4
0	0	0	0	1	1

Writing a 4+4 bits instruction:



RS	R/!W	DB7	DB6	DB5	DB4
0	0	0	0	1	1

RS	R/!W	DB7	DB6	DB5	DB4
0	0	0	1	1	1

Initialization stage

Instructions on page 24 of the Hitachi HD44780U datasheet

Power the display*

*The first line of the display is painted black.

Wait > 100 ms

RS	R/!W	DB7	DB6	DB5	DB4
0	0	0	0	1	1

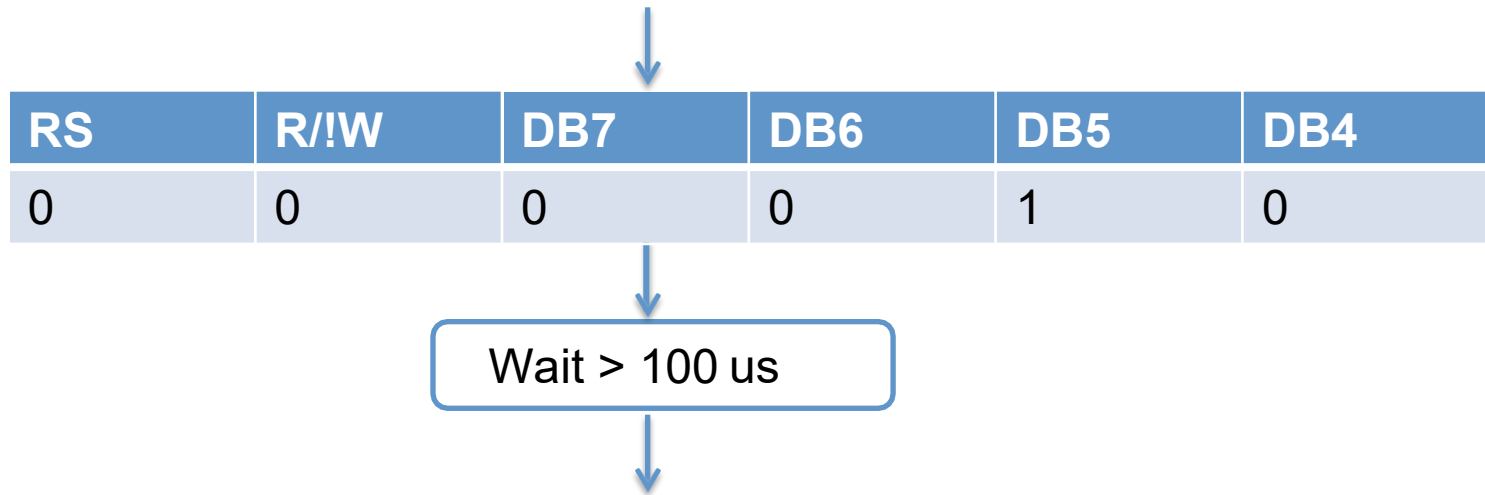
Wait > 4,1ms

RS	R/!W	DB7	DB6	DB5	DB4
0	0	0	0	1	1

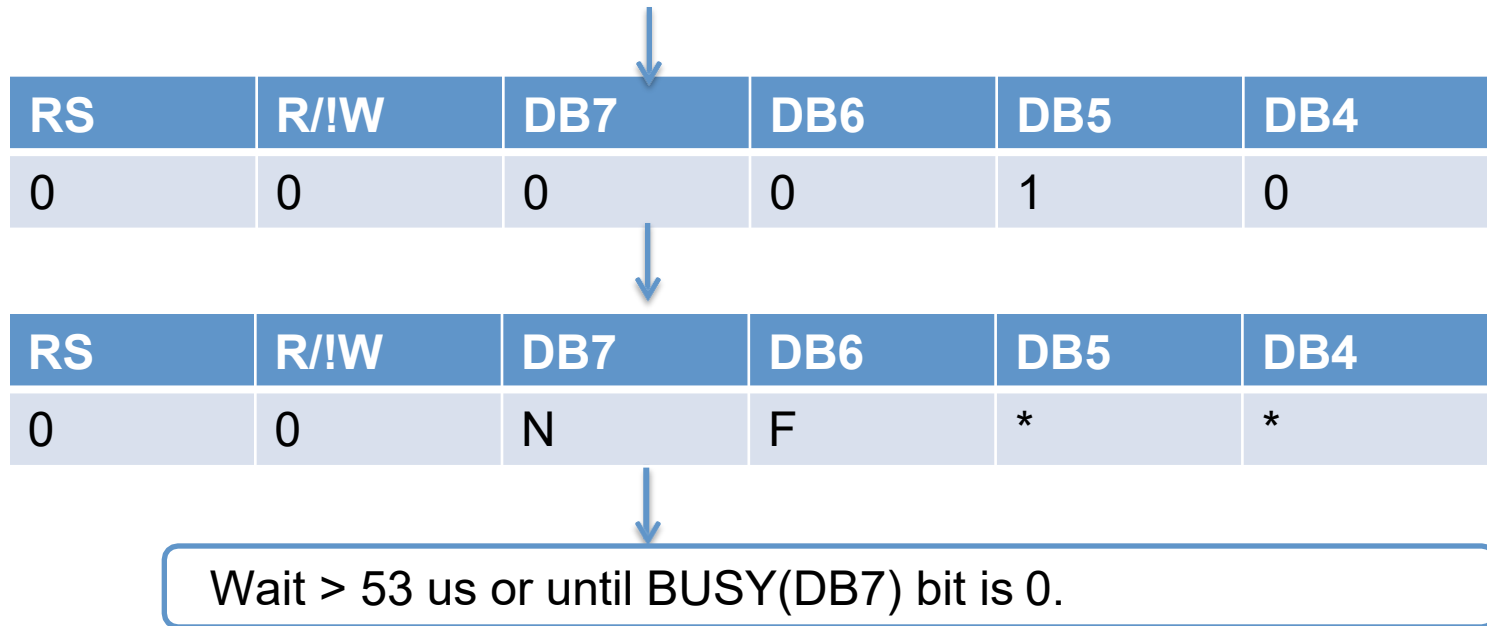
Wait > 100 us

RS	R/!W	DB7	DB6	DB5	DB4
0	0	0	0	1	1

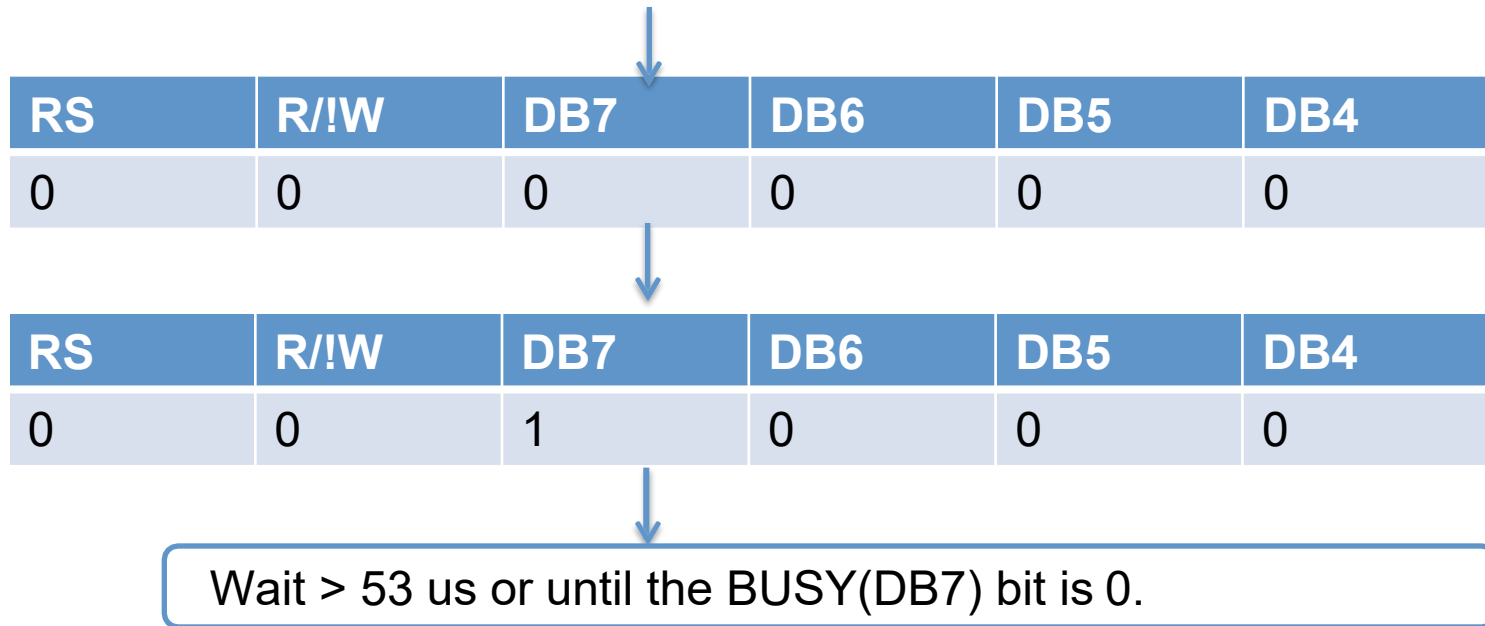
Wait > 100 us



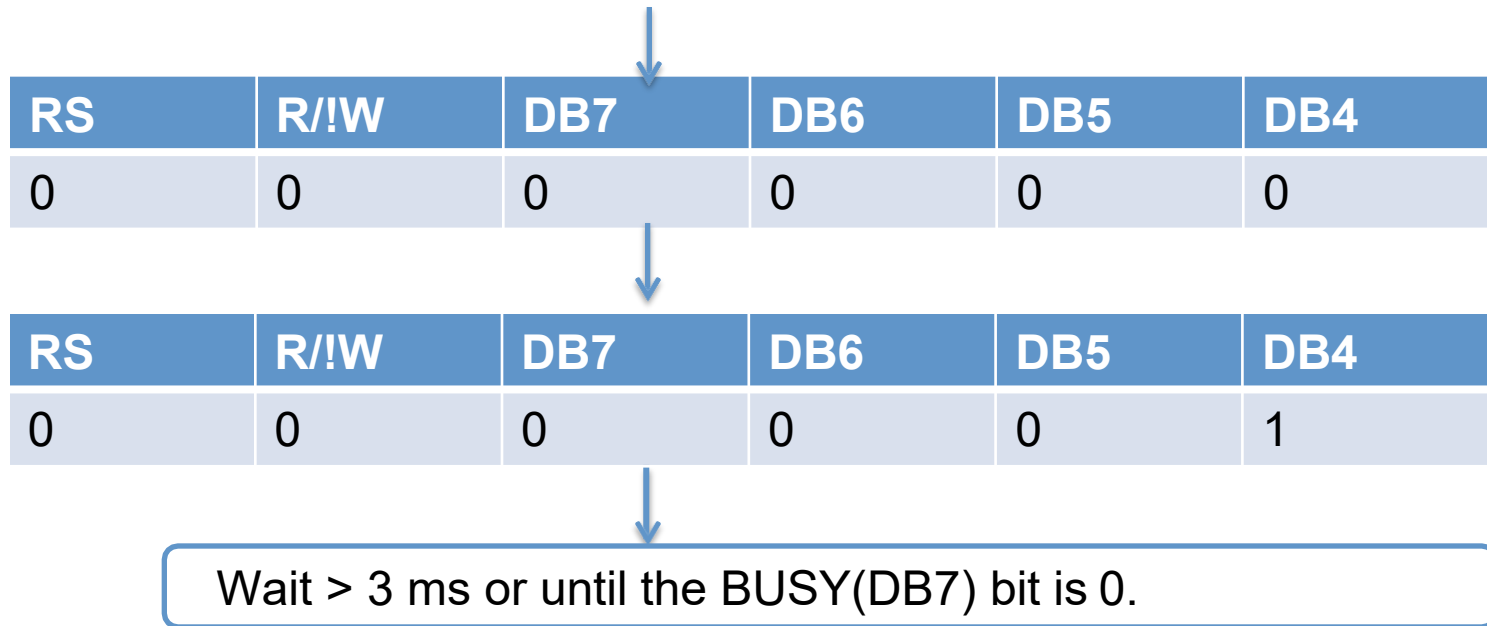
- From this moment, we are in 4-bits mode.
- When we activate the Enable, the LCD will get the Highest part of the instruction from DB[7..4] and will store it. Then, when we activate the Enable again the LCD will get the Lowest part of the instruction, will generate the 8-bits complete instruction and will execute it.



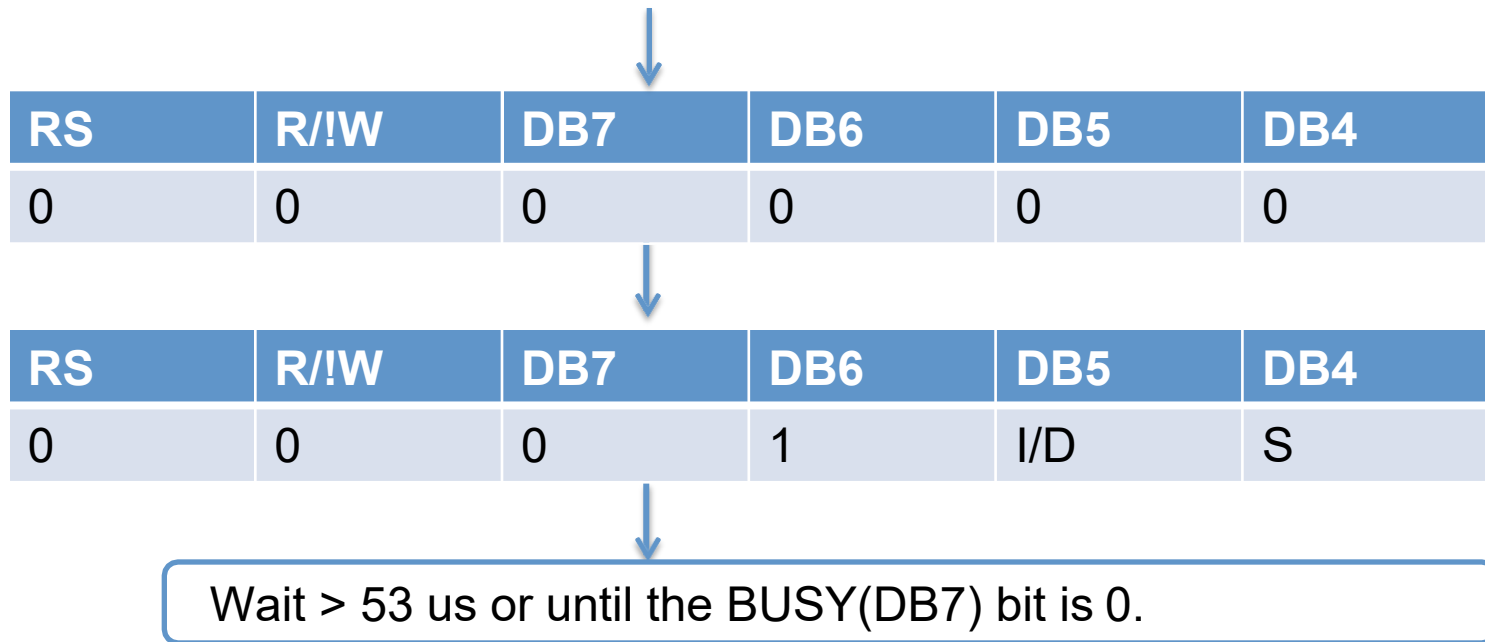
- N = Number of logical rows. In our case, N = 1.
- F = Resolution of each character. We will clear this bit because we want dimensions of 5x7.



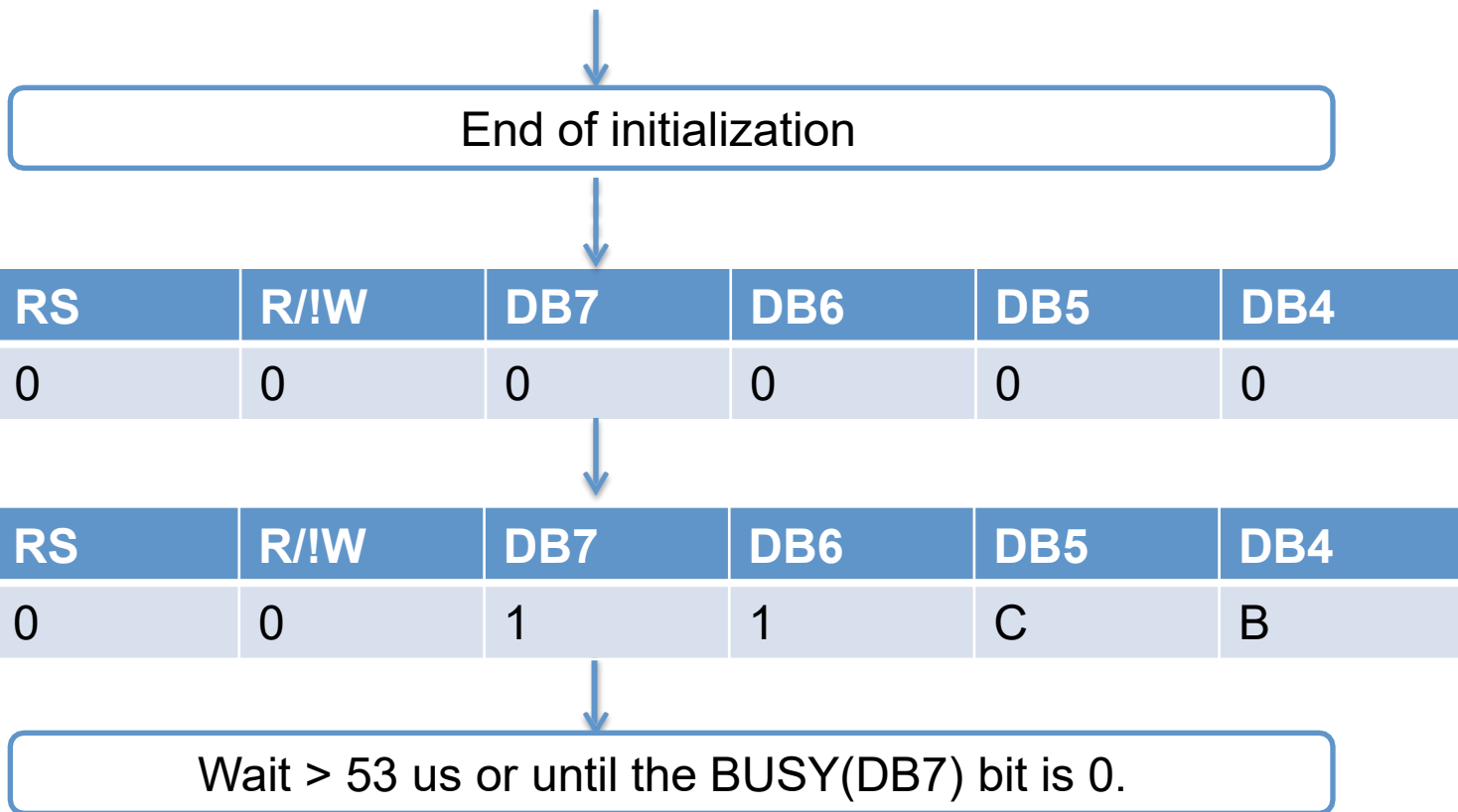
- From this moment, the first row of the LCD is cleared (turned off).
- Display On/Off control-> in this case, we set it to OFF to initialize it (DB7 = 1 and DB6= 0).



- “Clear Display” Instruction.
- Even when respecting all the indications of the datasheet, in some displays a wait of 3ms is necessary independently of the busy bit.



- This instruction determines how the cursor will move.
- To move from left to right (normal movement of the cursor) I/D=1.
- S is used to shift. Usually, S=0.



- Turn on the display (DB6=1).
- C is used to make the cursor visible (C=1) or invisible (C=0).
- B is the blinking bit. If B=1 Blinking is ON.
- **Repeat all the initialization process 3 times.**

Display characters using the LCD

Step 1: Select the position of the cursor (instruction)

Wait until the BUSY(DB7) bit is 0.



RS	R/!W	DB7	DB6	DB5	DB4
0	0	1	@DDRAM	@DDRAM	@DDRAM



RS	R/!W	DB7	DB6	DB5	DB4
0	0	@DDRAM	@DDRAM	@DDRAM	@DDRAM

Display
position

DDRAM
address

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
40	41	42	43	44	45	46	47	48	49	4A	4B	4C	4D	4E	4F

Step 2: Write an ASCII character to the LCD. (data)

Wait until the Busy (DB7) bit is 0.



RS	R/!W	DB7	DB6	DB5	DB4
1	0	ASCII [7]	ASCII [6]	ASCII [5]	ASCII [4]



RS	R/!W	DB7	DB6	DB5	DB4
1	0	ASCII [3]	ASCII [2]	ASCII [1]	ASCII [0]

Extra information

- **Datasheet Hitachi HD4478U:**

1. Page. 24: Instructions of the LCD.
2. Page. 12: Relation between the address of the DDRAM and the position of the cursor.
3. Page. 58: Timing diagrams.